

## 1. Project Overview

Face Seeker is a 100% offline, air-gapped computer vision desktop application built for law enforcement and forensic investigators. It automates the task of scanning hours of surveillance video footage to locate specific target subjects, replacing manual video review with high-speed deep learning inference.

## 2. How It Works (Internal Working Mechanism)

### Step 1: Target Ingestion & Vector Extraction

When a target photo is loaded, YuNet detects the face and SFace crops, aligns, and extracts a unique 128-dimensional floating-point feature vector representing the face structure.

### Step 2: Video Frame Decoding & Dual Resolution Pipeline

Video frames are decoded frame-by-frame. Frames are downscaled to 640x360 for high-speed YuNet detection (achieving 35-50+ FPS). Bounding box coordinates are then mathematically scaled back to full 1080p resolution.

### Step 3: Cosine Similarity Matching

SFace extracts a 128D vector for every face detected in the video frame. The engine computes the Cosine Similarity score against all loaded target vectors. If score >= Threshold (default 0.36), a match event is registered.

### Step 4: Asynchronous Multi-Threading & Queue Dispatch

AI scanning runs on a dedicated background worker thread to prevent UI freezing. Frame updates, progress percentages, FPS, and match payloads stream asynchronously to the main GUI thread via a thread-safe Queue.

### Step 5: High-Res Annotation & Report Generation

When a match is confirmed, the full 1080p frame is captured with bounding box coordinates and saved for side-by-side inspection, interactive seekbar plotting, and PDF incident report generation.

## 3. Complete Technology Stack

| Component                 | Technology                       |
|---------------------------|----------------------------------|
| Core Language             | Python 3.14                      |
| Face Detection AI Model   | YuNet ONNX (OpenCV)              |
| Face Recognition AI Model | SFace ONNX (OpenCV)              |
| Math & Matrix Engine      | NumPy                            |
| GUI Desktop Framework     | CustomTkinter (Win11 Dark Theme) |
| Image Processing Library  | Pillow (PIL)                     |
| PDF Report Generator      | FPDF2                            |
| Standalone Packaging      | PyInstaller 6.x                  |

## 4. Why Face Seeker is Better (Comparison with Existing Solutions)

| Dimension          | Typical GitHub Tools         | Enterprise Software       | Face Seeker (Our Solution)        |
|--------------------|------------------------------|---------------------------|-----------------------------------|
| Deployment         | Complex Python/dlib setup    | Expensive server contract | Single Portable Executable (.exe) |
| Model Overhead     | 2GB - 5GB PyTorch/dlib       | Proprietary cloud models  | 85KB + 37MB ONNX (Lightweight)    |
| Privacy & Security | Variable (Some use web APIs) | Cloud-connected servers   | 100% Air-Gapped & Offline (0 Net) |
| Processing Speed   | Slow (5 - 15 FPS CPU)        | Fast GPU Cluster          | 35 - 50+ FPS (Optimized YuNet)    |
| Forensic Reporting | Raw text output only         | Enterprise dashboards     | Side-by-side & PDF Reports        |